Physics Guards for Residual Policies at Breakaway: When Learned Corrections Should Defer to a Friction Model

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Code, data and protocol: https://github.com/CAN-Aidoo/breakaway-shield

Abstract

TBD

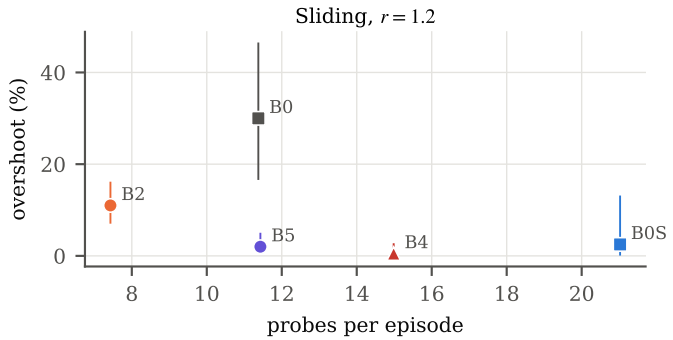


Figure 1. Measurements against safety at the largest static/kinetic ratio in the training range. The unshielded residual (B2) needs the fewest measurements but still overshoots. The shield-trained residual (B5) keeps overshoot near zero at about the model controller's measurement cost. The shielded model controller (B0S) pays far more measurements for the same safety. Bars: 95% Clopper–Pearson intervals.

I. Introduction

Residual learning keeps a hand-designed controller and learns a correction to its output [1, 2]. In earlier work we applied it to gravitational pivoting with a parallel-jaw gripper that senses the object only through wrist motor current [3]. The residual roughly halved the number of slow angle measurements per pivot, from 18.0 to 9.3, but it overshot targets more often. Almost every overshoot happened on one step, the step on which the rod first slipped [4]. A hand-made guard repaired most of the damage, but it was designed after we had seen the test results and carried no guarantee.

That failure has a physical cause. When a grasped object breaks away, friction falls from its static to its kinetic level, and the surplus of gravity over kinetic friction becomes kinetic energy. A quasi-static friction model, which both the base controller and our earlier guard rely on, ignores this energy. A learned correction that makes releases larger or faster than the model's falls exactly into the region where the model is least reliable.

This paper asks when a learned correction should defer to a friction model. We turn the first-slip guard into a *shield* [5]: a closed-form set of grip actions that keeps the post-breakaway travel within the target band whenever the true static-to-kinetic ratio $r = \mu_s/\mu_k$ does not exceed an assumed bound $r_{\max}$. Outside that set, the learned correction is dropped. We study two tasks, gravitational pivoting and controlled sliding of a rod through the fingers, in MuJoCo simulation with explicit stick–slip. Our contributions are:

- A breakaway shield for pivoting with a one-line closed form (Proposition 1), together with the exact conditions under which it reduces to our earlier quasi-static guard, and a timing shield for sliding (Proposition 2). Sliding needs a bound on time below the arrest force, not a force floor.
- A feasibility analysis showing that, from rest, a certified release toward a target θ^* exists only if $r < \varphi_{\lim}/\sin\varphi_{\lim}$. Shallow pivots are therefore infeasible for realistic r, and the shield's own infeasible region coincides with this physical one.
- A pre-registered simulation study (**Q1**–**Q4**) comparing the model controller, unshielded and shielded residual policies, and shields applied at training or only at test time, across r and under $r_{\max}$ mismatch, with certificate-violation rates reported alongside task metrics.

II. Related Work

Residual learning and physics priors. Residual RL learns an additive correction on top of a controller [1, 2] or a learned base policy [6]. Corrections are usually limited by a fixed magnitude or a penalty: ForceRFT caps a wrist-force-conditioned residual at fixed per-axis bounds [7]. Chen, Lu, Colgate and Lynch embed grasp-quality and fingertip-geometry priors into the reward and the hardware for in-hand rotation [8]. Our bound is different in kind: it is *state-dependent*, derived from contact mechanics, and enforced at run time.

Shields and safety filters. Shielding overrides unsafe actions using a model of the system [5]. Control-barrier-function filters [9, 10], Hamilton–Jacobi reachability [11] and predictive safety filters [12] generalise the idea; Brunke et al. review the field [13]. Adaptive shielding infers hidden parameters such as friction online [14], and ResSafe learns the safety filter itself as a residual [15]. Closest to us on the friction side, Johansson et al. filter bimanual manipulation actions with a Coulomb factor of safety evaluated by rollouts [16]. Closest in spirit, Shuto and

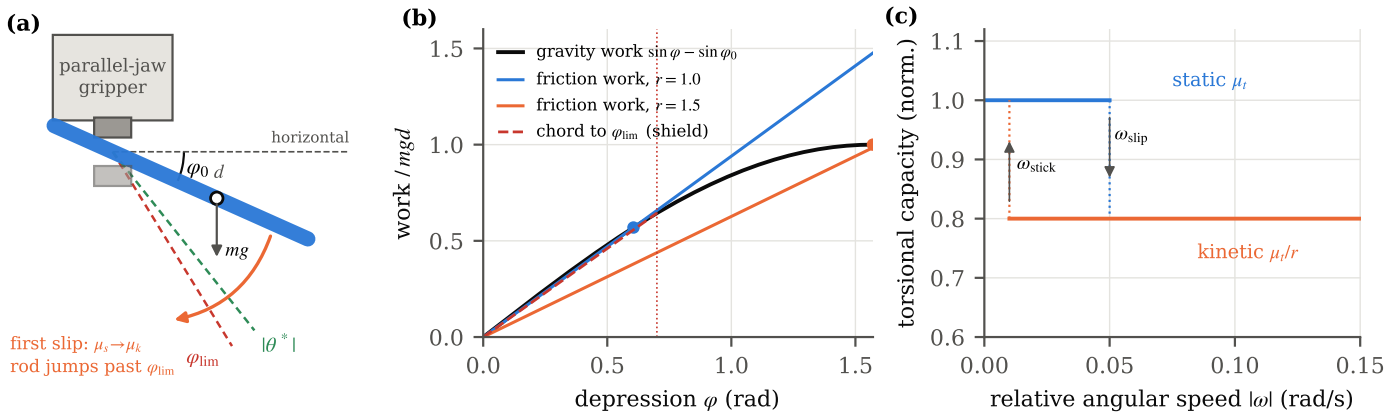


Figure 2. The breakaway problem. (a) A rod held off its centre of mass pivots under gravity; at the first slip friction drops from static to kinetic and the rod can run past the limit $\varphi_{\lim}$. (b) Energy balance from rest at $\varphi_0 = 0$: the rod stops where friction work (lines of slope $k = F/(rF_h)$) catches up with gravity work. With the same grip, $r = 1.5$ stops far later than $r = 1$. The shield requires the friction line to lie above the chord to $\varphi_{\lim}$. (c) The emulated friction law: static capacity until the relative speed exceeds ω_{slip}, kinetic capacity μ_t/r until it falls below ω_{stick}.

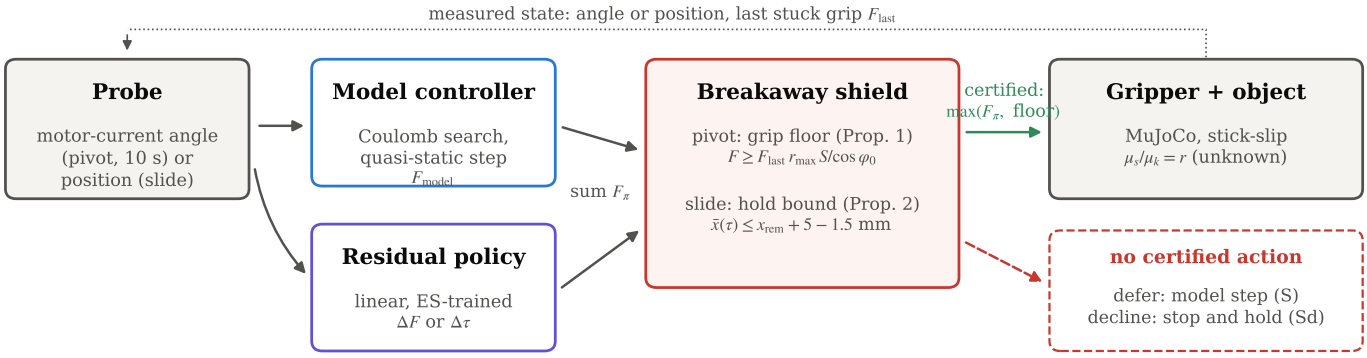


Figure 3. Shielded residual control. A probe returns the measured state and the last grip at which the object was stuck. The model controller proposes an action and the residual policy corrects it. The shield passes certified actions (clipping grip up, or the hold down, to the bound) and falls back to the model step (defer) or stops (decline) when no certified action exists.

Vargas derive a certified, sensorless grasp-damage bound from a stiffness lower bound and propose it as a safe-action parameter for learned grasping [17]. We make two changes. Our certificate rests on the static-to-kinetic *transition*, which a single-μ Coulomb model cannot express. And we actually train and test learners under it.

Pivoting and controlled sliding. Gravitational pivoting has been planned open loop [18], controlled adaptively [19] and learned with tactile sensing [20]. Dynamic in-hand sliding has been planned with soft-finger limit surfaces [21], and prehensile pushing uses Coulomb motion cones [22]. Soft-finger friction scales nonlinearly with load [23, 24]. Kolbert et al. found that MuJoCo under-predicted a measured pivot [25]. None of these works isolates breakaway as the failure mode of a learned correction.

III. Breakaway Shields

A. Pivoting

A rod of mass m is held at offset d from its centre of mass by two soft fingers with grip force F. Let $\varphi \in [0, \pi/2]$ be the depression of the centre of mass below horizontal. We assume (A1) a rigid rod rotating about the grip axis, (A2) torsional capacity linear in grip, $\tau_s = c_s F$ static and $\tau_k = c_s F/r$ kinetic, (A3) a commanded grip that is tracked and non-increasing during a release, (A4) no stored pad compliance and no damping, and (A5) gravity torque $mgd\cos\varphi$. The horizontal breakaway grip is $F_h = mgd/c_s$.

Proposition 1. *Suppose the rod is stuck at φ_0 under grip F_{last}, and the true ratio satisfies $r \leq r_{\max}$. Then any release to*

$$F \;\geq\; F_{\text{last}} \frac{r_{\max}}{\cos\varphi_0}\, S(\varphi_0, \varphi_{\lim}), \qquad S(a,b) = \frac{\sin b - \sin a}{b - a}, \quad (1)$$

leaves the rod stuck at some $\varphi \leq \varphi_{\lim}$.

Proof. Sticking at φ_0 implies $c_s F_{\text{last}} \geq mgd\cos\varphi_0$, so $F_h \leq F_{\text{last}}/\cos\varphi_0$. If the rod breaks away under final grip F, its kinetic energy obeys $\frac{1}{2} I\dot{\varphi}^2 \leq mgd\, g(\varphi)$ with $g(\varphi) = \sin\varphi - \sin\varphi_0 - k(\varphi - \varphi_0)$ and $k = F/(rF_h)$. By A3, the grip during sliding is at least F; by A4, no energy enters from compliance. The rod stops at the first root of g after φ_0. Because sin is concave on $[0, \pi/2]$, the chord slope $S(\varphi_0, \varphi)$ decreases in φ, so that

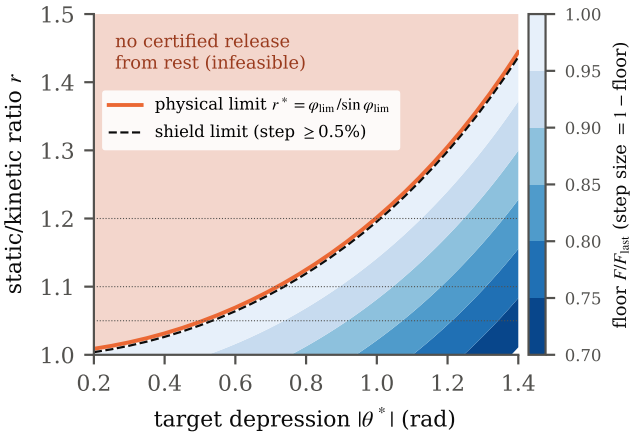


Figure 4. Where a certified pivot from rest exists. Shaded blue: the shield's grip floor as a fraction of F_{last} (smaller means larger steps). Pink: no certified release. The shield's limit (dashed) lies on the physical limit r^* (orange). For $r = 1.1$, targets shallower than about 0.75 rad are infeasible.

root is at most $\varphi_{\lim}$ iff $k \geq S(\varphi_0, \varphi_{\lim})$. At the root, $\cos\varphi < k$, so even kinetic friction exceeds gravity and the rod re-sticks. Substituting the bounds on F_h and r gives (1). □

Relation to the earlier guard. With $r_{\max} = 1$ and $\varphi_0 = 0$, (1) gives $F \geq F_{\text{last}} \sin\varphi_{\lim}/\varphi_{\lim}$. Our earlier guard, $F \geq F_{\text{last}} \cos\varphi_{\lim}$ [4], follows only if the energy balance is replaced by quasi-static equilibrium ($\cos\varphi = k$). The two floors differ by $\tan\varphi_{\lim}/\varphi_{\lim}$: about 1.03 at 0.3 rad, 1.20 at 0.7 rad and 1.56 at 1.0 rad. The quasi-static guard therefore ignores the inertia of the first slip even when $r = 1$.

Feasibility. A release that causes motion needs $F < F_h \cos\varphi_0$, while the shield needs $F \geq r F_h S$. A certified release from rest therefore exists only if

$$r < r^*(\varphi_{\lim}) = \varphi_{\lim}/\sin\varphi_{\lim}, \tag{2}$$

which is 1.04 at 0.5 rad, 1.09 at 0.7 rad and 1.19 at 1.0 rad. Because the shield computes its floor from F_{last} rather than the unknown F_h, it can only shrink the step toward breakaway, by the factor $r_{\max} S/\cos\varphi_0$. Whenever that factor is below one, the search still reaches breakaway, so the shield's infeasible region is the physical one (Fig. 4). The price is smaller steps, and so more measurements.

Deferral rule. At each step we compute the floor. If the floor is below $0.995 F_{\text{last}}$, the step is *certified* and the commanded grip is $\max(F_{\text{policy}}, \text{floor})$, a clip that never lowers grip. Otherwise no certified release exists. We pre-specified two fallbacks: *defer*, which drops the learned residual and takes the model controller's step (uncertified); and *decline*, which stops and holds. We set $\varphi_{\lim} = |\theta^*| + 0.05 - 0.02$ rad, the overshoot threshold minus a 2.5σ sensing margin.

B. Controlled sliding

A vertical rod held at its centre of mass slides along its axis under gravity. Static breakaway occurs at $F_b = mg/(2\mu_s)$, but stopping a sliding rod needs $F > rF_b$. Gravity does not weaken as the rod slides, so for $r > 1$ any release that causes breakaway accelerates the rod until the grip rises above rF_b. A force floor cannot bound this travel. What bounds it is the time spent below the arrest force.

Proposition 2. *Suppose the rod was last observed stuck at grip F_{stuck} and $r \leq r_{\max}$. For a planned grip profile $F(t)$ starting from rest, the travel is at most $\bar{x} = \int_0^T \bar{v}\,dt$, where $\dot{\bar{v}} = g\,(1 - F(t)/(r_{\max} F_{\text{stuck}}))$ and $\bar{v}$ is clipped at zero.*

Proof. While sliding, the acceleration is $g(1 - F/(rF_b))$, and $rF_b \leq r_{\max} F_{\text{stuck}}$ because the rod was stuck at F_{stuck}. So the true acceleration never exceeds $\dot{\bar{v}}$, whether positive or negative, and the true velocity never exceeds $\bar{v}$. □

The shield shortens the hold of each pulse, by bisection, until $\bar{x}$ is within the remaining distance plus 5 mm minus a 1.5 mm sensing margin. Unlike pivoting, sliding has no infeasible region caused by r. Infeasibility arises only while F_{stuck} is still loose, early in the search.

IV. Experimental Setup

Simulator. We use MuJoCo 3.15.0 [26] with the rig of [3, 4]: soft-finger contacts (condim 4), an elliptic cone, impratio 10, 20 noslip iterations and a 1 ms step. MuJoCo has a single friction coefficient per geom and no true stick state [27]. We therefore emulate stick–slip by switching the torsional coefficient (pivoting) or the sliding coefficient (sliding) of all contact geoms between μ and μ/r, with hysteresis on the relative speed: switching to kinetic friction above 0.05 rad/s or 5 mm/s, and back to static below 0.01 rad/s or 1 mm/s (Fig. 2c). With $r = 1$ the rig reproduces our earlier results exactly, to 10^{-6} rad. Section A reports the validation checks.

Tasks and controllers. *Pivoting* uses the object distribution and model controller of [4]: a $\times 0.94$ grip search until motion, then quasi-static steps toward the target; angle measurements have $\sigma = 0.008$ rad. *Sliding* lowers the rod by $x^* \in [10, 50]$ mm using release pulses. Each pulse releases in 0.1 s, holds for τ and re-clamps in 0.3 s, followed by a position measurement ($\sigma = 0.5$ mm). The model controller searches with $\times 0.9$ steps and then chooses the hold from a Coulomb ($r = 1$) prediction, corrected by the ratio of observed to predicted travel. In both tasks, each measurement is one *probe*, the costly sensing action.

Learning. Linear residual policies, as in [4], are trained with antithetic evolution strategies (1560 episodes per run). Training randomises $r \sim U[1, 1.1]$ for pivoting, with shield $r_{\max} = 1.1$, and $r \sim U[1, 1.2]$ for sliding, with $r_{\max} = 1.2$. For each task we train two methods with 5 seeds each: an unshielded residual, and a residual with the shield in the training loop (defer fallback).

Methods. B0 is the model controller; B1 adds the earlier quasi-static guard (pivoting only); B0S is the model controller with the shield. B2 is the unshielded residual; B3 adds the earlier guard (pivoting only). B4 applies the shield to the unshielded residual at test time only. B5 is the shield-trained residual. A suffix "d" denotes the decline fallback.

Table 1. Simulator validation (validation objects only; recorded before any test data).

Check	Quantity	Result
V1	slow-ramp breakaway / nominal	1.003–1.027
V2	intrinsic static/kinetic ratio	1.008 (1.006–1.010)
V2	emulated $r = 1.2$, measured	1.206
V3	creep at 1.05 F_h	$< 10^{-5}$ rad/s
V4	first-slip travel, dt halved	$\lvert\Delta\rvert < 0.025$ rad
V5	thresholds $\times 0.5$, $\times 2$	$\lvert\Delta\rvert < 0.01$ rad
V6	travel / Prop. 1 bound	0.90–0.97 (mean)
V6	releases over bound	24/128, max 0.048 rad

Protocol and statistics. The protocol, including hypotheses, sets, metrics and tests, was committed before any test evaluation; the sliding task was added as a dated deviation before any sliding test data existed. Each test set has 40 held-out objects evaluated at fixed r values, and a confirmation set of 40 fresh objects was run once after the analysis. Learned methods are averaged over seeds per object. We use paired Wilcoxon tests against B0, hierarchical bootstrap 95% CIs (seeds, then objects), Holm correction within each task's B5-vs-B0 family, and Clopper–Pearson intervals for rates.

V. Results

A. Simulator validation

Before any test run we checked the stick–slip emulation (protocol checks V1–V6, Table 1). Slow-ramp breakaway occurred within 0.3–2.7% of the nominal grip, for both $r = 1$ and $r = 1.2$. The contact itself has a small static/kinetic gap: breakaway at 0.988 and sliding at 0.980 of nominal capacity, an intrinsic ratio of 1.008, so an emulated $r = 1.2$ measured 1.206. Creep in the stuck phase was below 10^{-5} rad/s. Halving the time step changed first-slip travel by less than 0.025 rad, and scaling both switching thresholds by 0.5 or 2 changed it by less than 0.01 rad.

The direct test of Proposition 1 held the rod at 1.01 times its measured breakaway grip and then released it to qF_{last} (Fig. 5). Over 128 releases, simulated travel averaged 0.90–0.97 of the bound, so the bound is tight. However, 24 releases exceeded it, by at most 0.048 rad, more than our 0.02 rad sensing margin. We recorded this before testing (deviation D2) and expected certificate violations in the test runs.

B. Q1: Does the shield keep the residual's savings?

Sliding (Table 2). The unshielded residual B2 needs far fewer measurements than the model controller: 7.4 against 11.4 at $r = 1.2$, and 3.5–5.3 fewer at every r. The shield-trained residual B5 keeps only part of this. It saves -2.5 probes at $r = 1.0$ (95% CI -4.8 to -0.6, Holm $p = 0.020$) and -1.6 at $r = 1.1$ (not significant), and none at $r = 1.2$ or 1.3. In sliding, then, the safety bought by the shield costs most of the residual's measurement saving. Two comparisons show where the cost goes. Shielding the unshielded residual only at test time (B4) needs 13.1–15.0 probes, against 10.8–12.1 for B5: a policy trained under the shield learns to work within it. Shielding the model controller (B0S) nearly doubles its probes, to 21–27. The shield shortens its holds, so the rod stops short and needs more pulses.

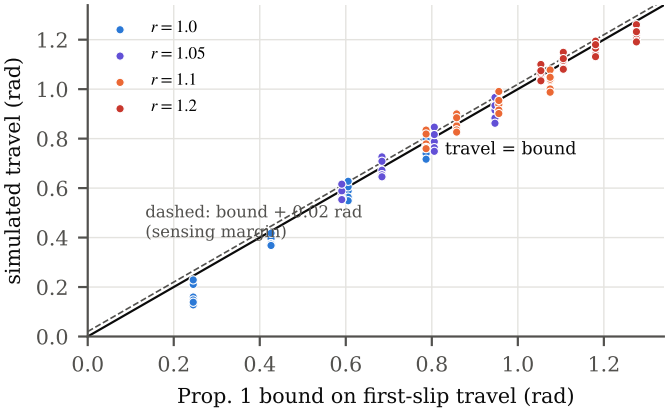


Figure 5. Proposition 1 against simulation (V6). Each point is one release from a measured stuck grip. Points above the solid line exceed the bound; points above the dashed line exceed it by more than the sensing margin.

C. Q2: Does the shield bound overshoot?

Sliding. The model controller overshoots 15, 8, 30 and 28% of episodes at $r = 1.0$, 1.1, 1.2 and 1.3. The Coulomb prediction it uses for the hold underestimates the post-breakaway acceleration once $r > 1$, and its travel correction adapts too late. The unshielded residual lowers this to 8–13%. The shield-trained residual overshoots 1–2.5% at every r, with 96–98% of episodes ending within 5 mm. Against B0, the reduction is -28 points at $r = 1.2$ (CI -42 to -15, Holm $p = 0.008$) and -25 at $r = 1.3$ (Holm $p = 0.015$). At $r \leq 1.1$, where B0 overshoots less, the reduction is not significant after correction.

The certificate itself, that a certified pulse travels no further than its limit, failed on 4, 3, 5 and 6 of about 1 900 certified B5 pulses per r (0.2–0.3%). Section E explains these failures.

D. Q3: How much of the task admits a certified action?

Sliding behaves as Proposition 2 predicts. Every shielded episode deferred exactly once, on its first pulse, while F_{stuck} was still unknown. No later pulse was infeasible, at any r (B0S 160 of 160 episodes, B4 and B5 800 of 800 each).

E. Q4: Why do violations and overshoots occur?

Attribution. We replayed every test episode with a certificate violation and classified each violating step (an exploratory analysis, run after the pre-registered tests). For sliding, 10 of the 32 violating steps occurred at $r = 1.3 > r_{\max}$, where no guarantee applies. One followed a recorded stuck grip that was below the true arrest requirement: the rod had slipped by less than the 2.5 mm motion threshold. The remaining 21 occurred with every assumption of Proposition 2 satisfied. All 21 were on three objects, all heavy ($m = 0.178$–0.192 kg). Direct probes on these objects found simulated travel up to 1.8 times the bound for short pulses released close to the stuck grip, where the bound is only a few millimetres. On light objects, travel stayed below the bound. We attribute these to MuJoCo's regularised friction, which lets a heavily loaded contact creep [27]; this is a simula-

Table 2. Sliding test set (40 objects; learned methods: 5 seeds, 200 episodes per cell). Within: final error ≤5 mm and held. Over: travel past the target by more than 5 mm. Probes: position measurements per episode. Shield $r_{\max} = 1.2$.

		$r = 1.0$			$r = 1.1$			$r = 1.2$			$r = 1.3$		
ID	method	within	over	probes	within	over	probes	within	over	probes	within	over	probes
B0	model	82	15	14.7	90	8	13.5	70	30	11.4	72	28	11.5
B0S	model + shield	78	**0**	24.2	65	2	26.6	75	2	21.0	68	5	21.1
B2	residual	90	8	9.3	90	10	8.2	89	11	7.4	86	13	7.9
B4	residual, shield at test	94	2	13.9	90	2	14.5	92	**0**	15.0	94	**1**	13.1
B5	shield-trained residual	98	2	12.1	98	**1**	11.9	96	2	11.4	96	2	10.8

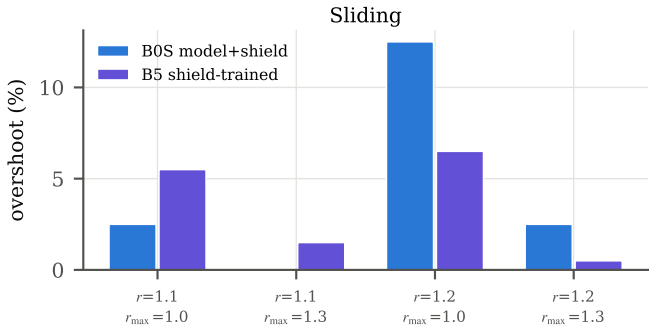


Figure 6. Overshoot under a wrong $r_{\max}$, for the shielded model controller (B0S) and the shield-trained residual (B5). Training and test-set shields use $r_{\max} = 1.1$ (pivoting) and 1.2 (sliding).

tor effect and not a sensing error.

Mismatch. Fig. 6 shows what happens when $r_{\max}$ is wrong. In sliding, assuming $r_{\max} = 1.0$ when the true r is 1.1 or 1.2 raises B5's overshoot to 6 and 6% and its violation rate to about 1%. Its probes fall to 8.5, the level of the unshielded residual. Assuming $r_{\max} = 1.3$ keeps overshoot at 2–0% at a cost of 13.3 probes. The bound trades measurements for overshoot monotonically, and an underestimate degrades the shield gradually rather than abruptly.

Confirmation. The confirmation sets, 40 fresh objects per task with r drawn from the training range, were run once after the test analysis had been committed. In sliding, B5 overshot 0% of episodes against 18% for B0 (Wilcoxon $p = 0.010$), ended within 5 mm in 98% against 82%, and used 11.9 probes against 14.0 (difference not significant). The test-set conclusions replicate.

VI. Limitations

Infeasible region. The main limitation is physical rather than algorithmic. By (2), no quasi-static release from rest can take the rod to a shallow target without passing it once r exceeds $\varphi_{\lim}/\sin\varphi_{\lim}$. For $r = 1.1$ this rules out every target shallower than about 0.75 rad. In this region the shield can only defer, which reproduces the model controller's overshoot, or decline, which stops short. A fast slip-triggered re-clamp (a "catch") could reach these targets: in a separate probe, a 50 ms detection-plus-clamp delay limited the first-slip jump to 0.03–0.07 rad even at $r = 1.5$. However, it needs slip detection far faster than our 10 s current-based angle measurement, and we leave it to future work.

Certified under a model, not under the world. Proposition 1 assumes a rigid rod, torsional capacity linear in grip, constant kinetic friction and no stored pad energy. In simulation the bound held on average, with simulated travel at 0.90–0.97 of the bound, but 24 of 128 direct test releases exceeded it by up to 0.048 rad. We attribute this to soft-contact geometry, and it exceeds our 0.02 rad sensing margin. On hardware, three effects could make the bound optimistic: soft-finger friction grows sublinearly with load [23], so capacity falls faster than grip when grip is reduced; elastomer pads store torsional energy before breakaway; and static friction rises with dwell time [28]. A practical value of $r_{\max}$ should therefore be measured on the actual pad–object pair, at the hold times the controller uses, and taken from the upper tail of the measured ratios.

Simulator. MuJoCo has no true stick state [27]. We emulated static and kinetic friction by switching coefficients with hysteresis. The emulation reproduced the intended ratio (an emulated 1.2 measured 1.206) and an intrinsic ratio of 1.008, and first-slip travel was insensitive to the time step and switching thresholds. It is nonetheless a step-function Stribeck curve without rate-and-state effects [29]. Kolbert et al. found that MuJoCo under-predicted a real pivot [25], so absolute travel values should not be transferred to hardware.

Learning and scope. We used linear policies, evolution strategies, 1560 episodes per run and 5 seeds per method. Larger policies or other optimisers could change the comparison between shield-in-training and shield-at-test. Sensing during training was a noise model, not the motor-current pipeline. Lift-with-slip, the third task we originally planned, is not studied.

VII. Conclusion

TBD

Acknowledgements and Disclosure

Simulation code, analysis and drafting were carried out with the assistance of an AI model (Claude, Anthropic) under the author's direction. The author takes responsibility for the content. All results are from simulation.

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